

Fall 1989

Problem 1 (Fa89). Let A be a finite abelian group, and m the maximum of the orders of the elements of A . Put $S = \{a \in A \mid |a| = m\}$. Prove that A is generated by S .

Problem 2 (Su78, Fa89). Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is continuous with $f(0) = 0$, and for $0 < x < 1$, f is differentiable and $0 \leq f'(x) \leq 2f(x)$. Prove that f is identically 0.

Problem 3 (Fa89). Let A be a real, upper triangular, $n \times n$ matrix that commutes with its transpose. Prove that A is diagonal.

Problem 4 (Sp87, Fa89). Let f be an analytic function in the open unit disc of the complex plane such that $|f(z)| \leq C/(1 - |z|)$ for all z in the disc, where C is a positive constant. Prove that

$$|f'(z)| \leq \frac{4C}{(1 - |z|)^2}$$

Problem 5 (Fa89). Let G be a group, G' its commutator subgroup, and N a normal subgroup of G . Suppose that N is cyclic. Prove that $gn = ng$ for all $g \in G'$ and all $n \in N$.

Problem 6 (Fa89). Let $X \subset \mathbb{R}^n$ be a closed set and r a fixed positive real number. Let $Y = \{y \in \mathbb{R}^n \mid \|x - y\| = r \text{ for some } x \in X\}$. Show that Y is closed.

Problem 7 (Fa89). Let A and B be diagonalizable linear transformations of $\mathbb{R}^n$ into itself such that $AB = BA$. Let E_λ be an eigenspace of A . Prove that the restriction of B to E_λ is diagonalizable.

Problem 8 (Su83, Sp84, Fa89). Compute $\int_0^\infty \frac{\log x}{x^2 + a^2} dx$ where $a > 0$ is a constant.

Problem 9 (Fa89). Let $\mathbb{F}$ be a field, $\mathbb{F}[x]$ the polynomial ring in one variable over $\mathbb{F}$, and R a subring of $\mathbb{F}[x]$ with $\mathbb{F} \subset R$. Prove that there exists a finite set $\{f_1, f_2, \dots, f_n\}$ of elements of $\mathbb{F}[x]$ such that $R = \mathbb{F}[f_1, f_2, \dots, f_n]$.

Problem 10 (Fa89, Sp94). Let α be a number in $(0, 1)$. Prove that any sequence (x_n) of real numbers satisfying the recurrence relation

$$x_{n+1} = \alpha x_n + (1 - \alpha)x_{n-1}$$

has a limit, and find an expression for the limit in terms of α , x_0 and x_1 .

Problem 11 (Fa89). Let φ be Euler's totient function; so if n is a positive integer, then $\varphi(n)$ is the number of integers m for which $1 \leq m \leq n$ and $\gcd\{n, m\} = 1$. Let a and k be two integers, with $a > 1, k > 0$. Prove that k divides $\varphi(a^k - 1)$.

Problem 12 (Fa89). Let $f(z)$ be analytic in the annulus $\Omega = \{1 < |z| < 2\}$. Assume that f has no zeros in Ω . Show that there exists an integer n and an analytic function g in Ω such that, for all $z \in \Omega$, $f(z) = z^n e^{g(z)}$.

Problem 13 (Su85, Fa89). Let A be an $n \times n$ real matrix and $A^\top$ its transpose. Show that $A^\top A$ and $A^\top$ have the same image.

Problem 14 (Fa89). Let $X \subset \mathbb{R}^n$ be compact and let $f : X \rightarrow \mathbb{R}$ be continuous. Given $\varepsilon > 0$, show there is an M such that for all $x, y \in X$,

$$|f(x) - f(y)| \leq M\|x - y\| + \varepsilon$$

Problem 15 (Fa89). Let G_n be the free group on n generators. Show that G_2 and G_3 are not isomorphic.

Problem 16 (Fa79, Fa89). Prove that the polynomial

$$p(z) = z^{47} - z^{23} + 2z^{11} - z^5 + 4z^2 + 1$$

has at least one zero in the disc $|z| < 1$.

Problem 17 (Sp80, Fa89). Show that a vector space over an infinite field cannot be the union of a finite number of proper subspaces.

Problem 18 (Fa89). Let the function $f : [0, 1] \rightarrow [0, 1]$ have the following properties:

- f is of class $\mathcal{C}^1$;
- $f(0) = f(1) = 0$;
- f' is nonincreasing, i.e., f is concave.

Prove that the arclength of the graph of f does not exceed 3.